

B.Tech Degree VI Semester Examination April 2012

ME 601 INSTRUMENTATION AND CONTROL SYSTEMS (2006 Scheme)

Time: 3 Hours

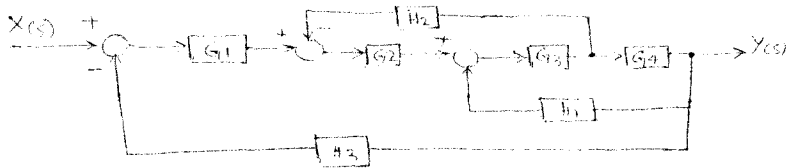
Maximum Marks: 100

PART A

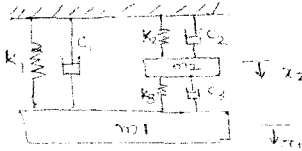
(Answer ALL questions)

(8 × 5 = 40)

- I. (a) Explain the terms state sensitivity and linearity of an instrument.
 (b) What is the impulse response of a first order instrument? Explain.
 (c) Explain the working of any one pyrometer.
 (d) Explain the working of a Gieger Muller counter.
 (e) Find the overall transfer function of the multiple loop feedback control system by block diagram reduction.



- (f) Find the system equation of the mechanical system shown. Also find the force-voltage analogy.



- (g) The characteristic equation of a system is given as $s^4 + 21s^3 + 36s + 20 = 0$. Find whether the system is stable or not.
 (h) Explain the working of a stepper motor.

PART B

(4 × 15 = 60)

- II. What are the various methods of connection for interfacing and modifying inputs to an instrument?

OR

- III. The discharge coefficient C_d of an orifice can be found by collecting water that flows through during a time interval when it is under constant head $C_d = \frac{W}{t_p A \sqrt{2gh}}$. Find C_d

and its possible error if $W = 385 \pm 0.2$ Kg, $A = \frac{\pi}{4} d^2$, $d = 0.013 \pm 0.000025$ m, $t = 600 \pm 2$ s, $g = 9.18 \pm 0.1\%$ m/s², $\rho = 1000 \pm 1\%$ Kg/m³, $h = 3.6 \pm 0.003$ m.

- IV. Explain the working of a strain rosette. How temperature compensation is done to strain Gauges?

OR

(P.T.O.)

V. Explain the working of the following with neat sketches.

- (i) ORSAT's apparatus
- (ii) Hydraulic dynamometer

VI. (a) Consider a system represented by the differential equation $\frac{d^2 y}{dt^2} + 4 \frac{dy}{dt} + 3y = 2r(t)$, where the initial conditions are $y(0) = 1$, $\frac{dy}{dt}(0) = 0$ and $r(t) = 1$, $t \geq 0$. What is the response of the system $y(t)$? Determine the steady state response.

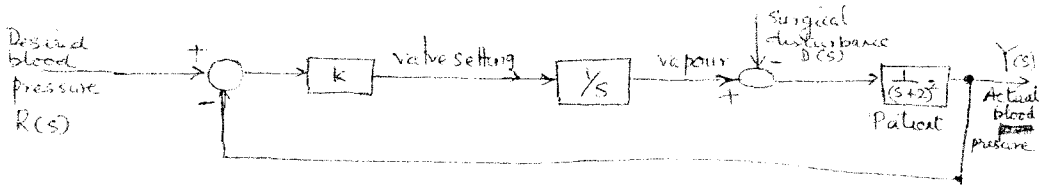
(b) A laser jet printer uses a laser beam to print copy rapidly. The laser is positioned by a control input $r(t)$ so that we have $Y(s) = \frac{5(s+100)}{s^2 + 60s + 500} \cdot R(s)$. The input $r(t)$ represents the desired position of the laser beam.

- (i) If $r(t)$ is a unit step input, find the output $y(t)$.
- (ii) What is the final value of $y(t)$?

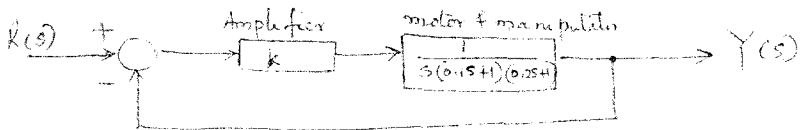
OR

VII. A system that controls the mean arterial pressure during anesthesia has been designed and tested. The level of arterial pressure is considered to be an indication of depth of anesthesia during surgery. A block diagram of the system is shown, where the impact of surgery is represented by the disturbance $D(s)$.

- (i) Determine the steady state error due to a disturbance $D(s) = \frac{1}{s}$
(Let $R(s) = 0$)
- (ii) Determine the steady state error for a ramp input $r(t) = t$, $t > 0$
(Let $D(s) = 0$).



VIII. The figure shows a manipulator control system used in a welding robot. Determine the amplifier gain k so that the steady state error for a ramp input $r(t) = At$ (where $A = 1$ mm/s) is less than or equal to 0.1mm, while a stable response is maintained. Plot the root locus.



OR

IX. A robot arm has a joint control open-loop transfer function $G(s) = \frac{2572}{(s+45.3)(s+341)}$. Plot the frequency response. (Magnitude and phase).